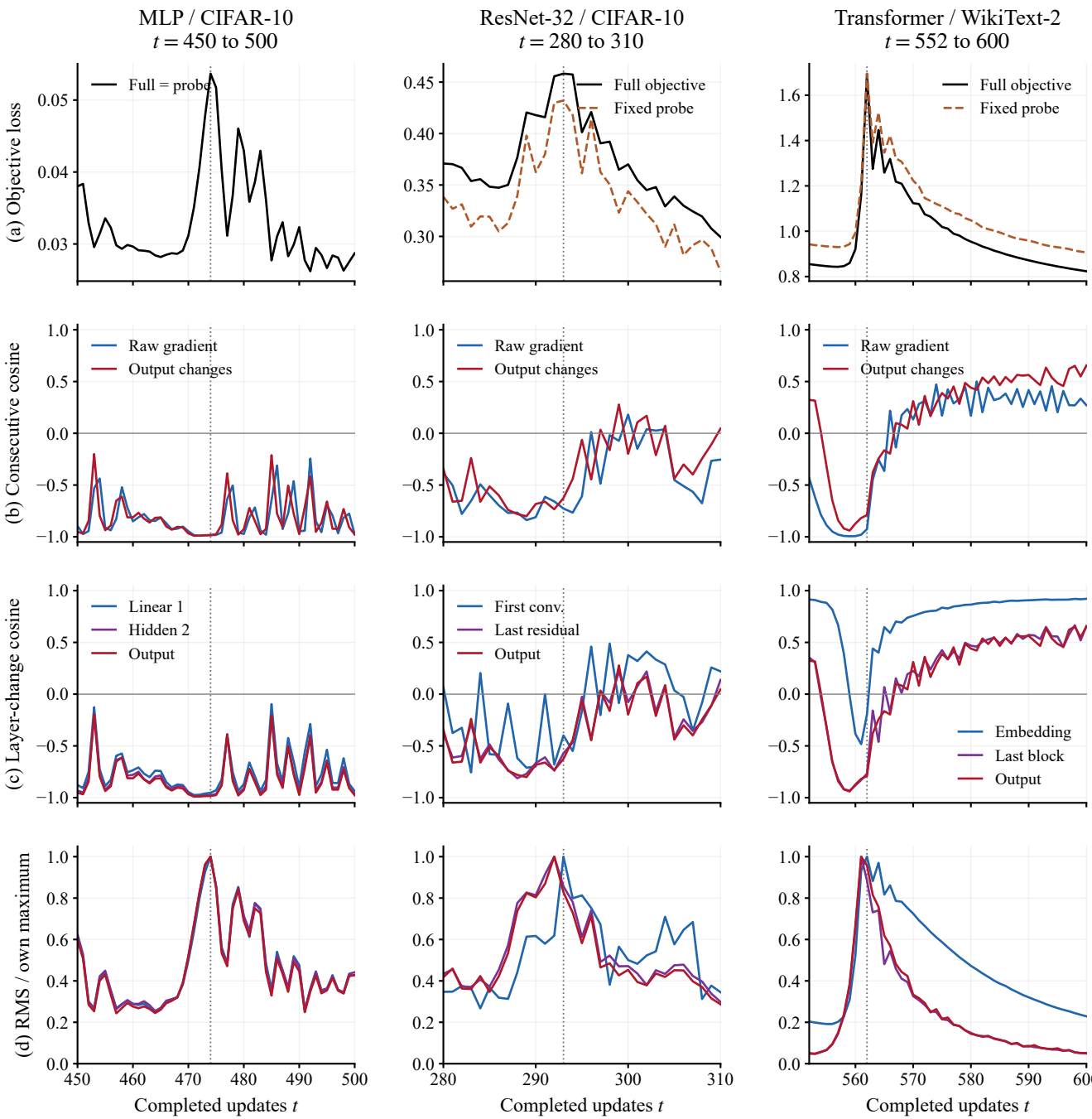

MLP / CIFAR-10
t = 450 to 500
ResNet-32 / CIFAR-10
t = 280 to 310
Transformer / WikiText-2
t = 552 to 600
(a) Objective loss
(b) Consecutive cosine
(c) Layer-change cosine
(d) RMS / own maximum
Full = probe
Full objective
Fixed probe
Raw gradient
Output changes
Linear 1
Hidden 2
Output
First conv.
Last residual
Output
Embedding
Last block
Output
Completed updates t
Completed updates t
Completed updates t